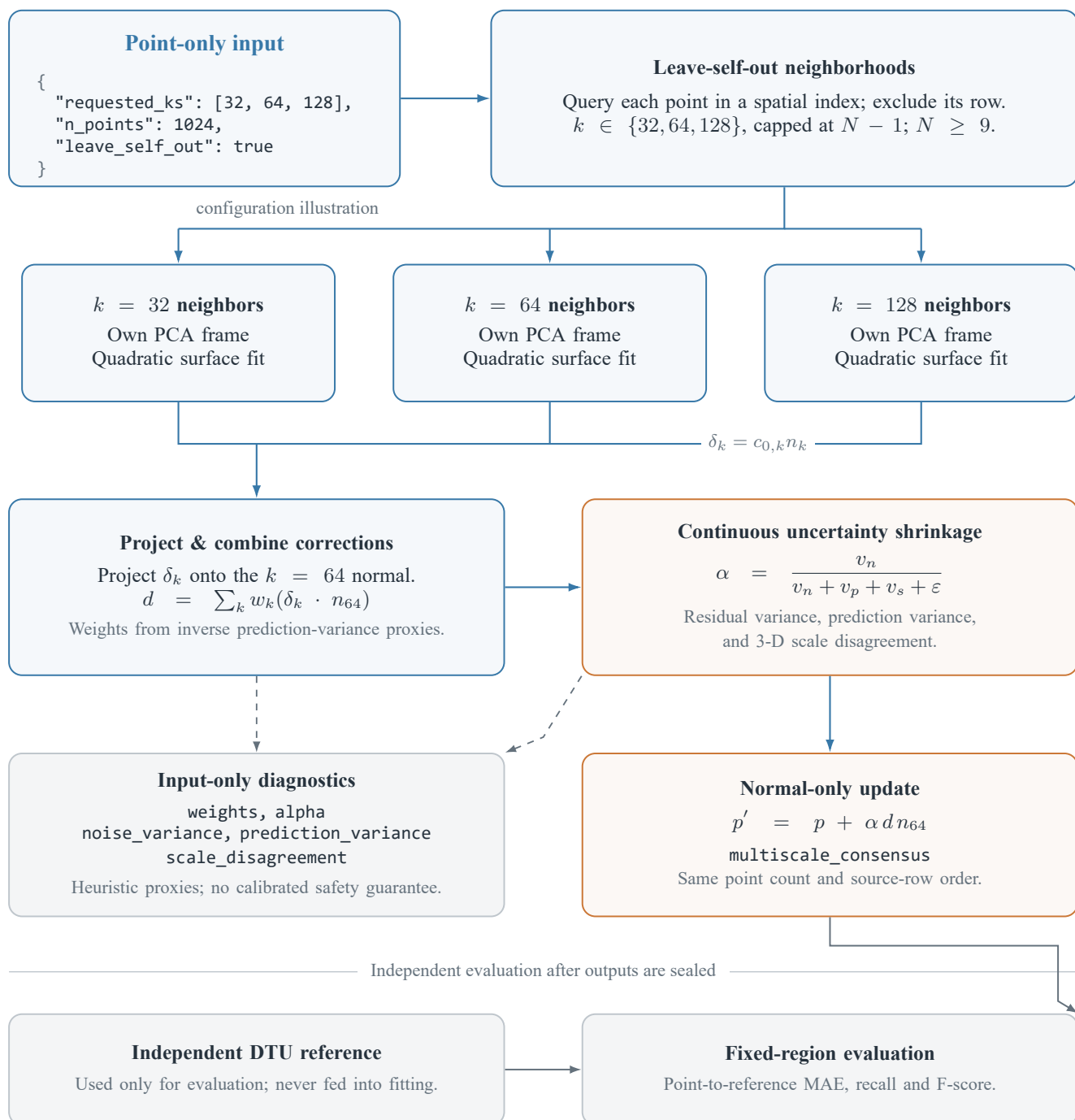


01 从局部曲面到连续阻尼

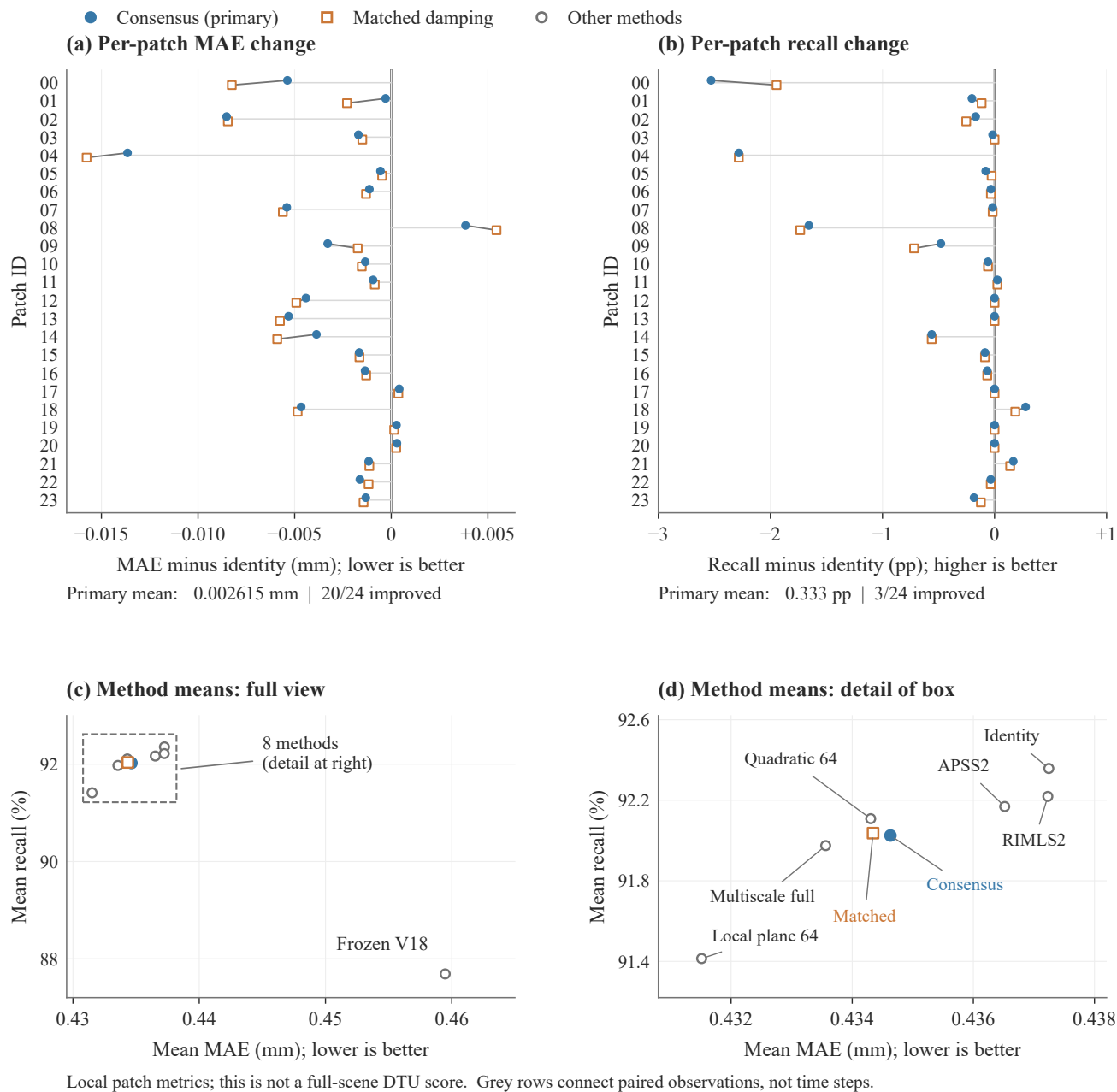


V22 为每个查询点建立排除自身的多尺度邻域，各自拟合局部二次曲面。不同尺度的修正投影到 $k=64$ 法向后融合，再施加连续阻尼。独立参考仅用于输出封存后的评分；该算法没有“拒绝即回退”的安全门控。

02 逐片区变化与几何覆盖权衡

Scan37 confirmation: local geometry and coverage

24 patches · 9 frozen methods · equal patch weights · recall threshold 1 mm



scan37 新确认场景的 24 个片区。逐片区变化均与同片区 identity 配对；主候选与相同整体位移 RMS 的常数阻尼同时展示。汇总均为片区等权均值，不能当作全场景官方 DTU 分数，也不代表独立场景数量为 24。

03 九种方法的完整数值对照

Scan37: geometry and coverage across nine methods

One new scene; 24 disjoint source-vertex patches, 1,024 input points per patch.

Method	MAE↓ mm	Recall↑ % at 1 mm	F-score↑ at 1 mm	Move RMS mm
Identity	0.437246	92.358	0.912854	0.000000
Frozen V18	0.459480	87.688	0.887500	0.376979
APSS2	0.436517	92.169	0.912036	0.034471
RIMLS2	0.437229	92.219	0.912017	0.030153
Local plane, $k = 64$	0.431515	91.414	0.908295	0.155598
Local quadratic, $k = 64$	0.434305	92.109	0.911920	0.097220
Multiscale, full update	0.433563	91.975	0.911316	0.088351
V22 consensus (primary)	0.434631	92.025	0.911513	0.060456
Matched-RMS constant damping	0.434344	92.037	0.911430	0.060456

Blue identifies the preselected primary method, not the best score. Every metric is averaged equally over the 24 patches. Evaluation uses the fixed observed local region; scored point counts can differ from the 1,024-point input. Recall measures reference coverage within 1 mm. These are local patch results, not official full-scene DTU scores.

Paired reading. Against identity, the primary method reduces mean MAE by 0.598%, while recall decreases by 0.333 percentage points. Matched-RMS constant damping has a slightly lower MAE than the primary method.

216 条已发布记录（24 片区 × 9 方法）直接重算。距离单位为 mm，召回率阈值为 1 mm。蓝色仅标识事前主候选。MAE 的轻微下降伴随召回和 F-score 代价，且主候选没有胜过同位移 RMS 常数阻尼。

04 真实代码与未通过的升级判据

V22: the implemented multiscale correction

Verbatim source excerpt: reconstruction_v22/operator.py, lines 119–138.

```

anchor = quadratic[control_k]
normals = anchor["normal"]
delta_vectors = np.stack([quadratic[k]["correction"][:, None] * quadratic[k]["normal"]
                           for k in actual_ks], axis=1)
projected = np.einsum("nsi,ni->ns", delta_vectors, normals)
point_variance = np.median(np.stack([quadratic[k]["noise_variance"] for k in actual_ks], axis=1), axis=1)
prediction_variances = np.stack([quadratic[k]["prediction_variance"] for k in actual_ks], axis=1)
# Numerical floor is dimensional and tiny; it is not a sensor-noise guess.
floor = np.maximum(anchor["tangent_scale"]**2 * 1e-24, np.finfo(float).tiny)
safe_variances = np.maximum(prediction_variances, floor[:, None])
# Equivalent inverse-variance ratios, without overflow for identical points.
precision = safe_variances.min(axis=1, keepdims=True) / safe_variances
weights = precision / precision.sum(axis=1, keepdims=True)
mean_delta = np.sum(weights * projected, axis=1)
mean_vector = mean_delta[:, None] * normals
disagreement = np.sum(weights * np.sum((delta_vectors - mean_vector[:, None])**2, axis=2), axis=1)
# Fits share observations; do not claim a 1/number_of_scales variance gain.
prediction_variance = np.sum(weights * prediction_variances, axis=1)
alpha = point_variance / (point_variance + prediction_variance + disagreement + floor)
consensus = points + (alpha * mean_delta[:, None] * normals

```

The applied displacement is $\alpha_i d_i \mathbf{n}_{i,64}$, where

$$\alpha_i = \frac{v_{n,i}}{v_{n,i} + v_{p,i} + v_{s,i} + \varepsilon_i}.$$

Shared observations are not treated as independent evidence: prediction variance is not divided by the number of scales. The factor α is heuristic damping, not a calibrated probability or a rejection gate.

Published comparison against identity SUMMARY.json / results.confirmation.comparisons

```

{
  "identity": {
    "paired": 24,
    "relative_gain": 0.005980615532293565,
    "win_fraction": 0.8333333333333334,
    "recall_difference": -0.003325902214863378,
    "passed": false
  }
}

```

passed: false means the predeclared composite improvement criterion was not met; it does not mean execution failed. Gains and recall differences in this JSON are fractions, not percentages.

上方原样摘录 operator.py 第 119 – 138 行，折行箭头只用于排版；下方来自已发布 SUMMARY.json 的实际比较。passed=false

表示未达到预设综合升级条件，并非程序执行失败。源码保留了“相关尺度不能虚构独立方差收益”的说明。